**Placeholder University**

**Placeholder Doctoral School**
*Information Sciences & Technologies*

# Doctoral Thesis

Presented by:
**Fourth Student**

# A Placeholder Title for a Placeholder Thesis

Defended publicly on **September 30th, 2026** before the jury composed of:

**Firstname Reviewerone**, Senior Researcher, HDR (Reviewer)
Placeholder Laboratory, Placeholder University

**Firstname Reviewertwo**, Professor, HDR (Reviewer)
Placeholder Institute

**Firstname Examinerone**, Researcher, HDR (Examiner)
Placeholder Institute

**Firstname Lastname**, Associate Professor, HDR (Supervisor)
Placeholder Laboratory

**Second Coauthor**, Researcher (Co-supervisor)
Placeholder Laboratory

# Table of Contents




**Acknowledgements** ..... **3**

**I Introduction** ..... **5**
- I.1 Context ..... 6
- I.2 Problem Statement ..... 6
- I.3 Outline ..... 6

**II State of the Art** ..... **7**
- II.1 Early Approaches ..... 8
- II.2 Modern Approaches ..... 8
- II.3 Summary ..... 8

**III A Placeholder Contribution** ..... **9**
- III.1 Method ..... 10
- III.2 Results ..... 10
- III.3 Discussion ..... 11

**IV Conclusion & Perspectives** ..... **13**
- IV.1 Summary ..... 14
- IV.2 Perspectives ..... 14

**A Proofs** ..... **15**
- A.1 Existence of a Placeholder Minimizer ..... 16
- A.2 A Proof Without a Statement ..... 16

**Bibliography** ..... **17**

**Acronyms** ..... **19**

# Acknowledgements



Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat.

Duis aute irure dolor in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt mollit anim id est laborum.

This chapter carries no number, and does not appear in the mini table of contents of the chapters that follow. It can still be referred to like any other one: `\nameref{phd_acknowledgements}` gives Acknowledgements.

# Chapter I

# Introduction

A PhD manuscript uses exactly the same commands as an HDR one: only the `phd` option of the document class differs.

This chapter introduces the placeholder problem addressed in this thesis, and outlines the rest of the document.


I.1 Context . . . . . 6
I.2 Problem Statement . . . . . 6
I.3 Outline . . . . . 6

## I.1 Context

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua [Ipsum *et al.*, 1998]. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat [Tempor *et al.*, 2009].

The rest of this document uses ML vocabulary, and in particular the Convolutional Neural Network (CNN). Note that CNNs is a separate entry of `acronyms.tex`, declared as a variation of `cnn`: its full name is Convolutional Neural Networks, it does not appear in the list of acronyms, and its link leads to the entry of the acronym it varies. Recall that State Of The Art (SOTA) results in this placeholder field are due to [Amet *et al.*, 2017].

## I.2 Problem Statement

**Problem 1** Placeholder problem

Given a set of placeholder observations $\{\boldsymbol{x}_i\}_{i=1}^N$ and their placeholder targets $\{\boldsymbol{y}_i\}_{i=1}^N$, find a placeholder model $f_\theta$ that generalizes to unseen placeholder observations.

Duis aute irure dolor in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla pariatur. Problem 1 is addressed in Chapter III, after the review of Chapter II.

## I.3 Outline

Chapter II reviews the placeholder literature. Chapter III presents our contribution, evaluated in Section III.2. Chapter IV concludes, and Appendix A gathers the proofs. The document opens on the Acknowledgements and closes on the Bibliography and the list of Acronyms: these three chapters are unnumbered, and are therefore recalled by their title rather than by a number.

# Chapter II

# State of the Art



This chapter reviews the placeholder literature, and positions the contribution of Chapter III with respect to it.


II.1 Early Approaches ..... 8
II.2 Modern Approaches ..... 8
II.3 Summary ..... 8

## II.1 Early Approaches

Lorem ipsum dolor sit amet, consectetur adipiscing elit [Ipsum *et al.*, 1998]. Sed do eiusmod tempor incididunt ut labore et dolore magna aliqua, as summarized in the textbook of [Tempor *et al.*, 2009].

**Definition 1** Placeholder representation

A *placeholder representation* of an observation $\boldsymbol{x}$ is any vector $\boldsymbol{z} = g(\boldsymbol{x})$ such that $\boldsymbol{z}$ carries no more information than $\boldsymbol{x}$.

## II.2 Modern Approaches

Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat [Amet *et al.*, 2017]. CNNs are now trained on a single GPU, which was not the case for the earlier approaches of Section II.1. Figure 1 summarizes the pipeline they all share.




Figure 1 An overview of the placeholder pipeline shared by modern approaches. The same figure is reused in Chapter III, which is perfectly fine as long as the labels differ.

## II.3 Summary

Table 1 compares the approaches discussed above. None of them addresses Problem 1 directly, which motivates the contribution of Chapter III.

| Approach | Placeholder | Scalable | Interpretable |
|---|---|---|---|
| [Ipsum *et al.*, 1998] | ✓ | ✗ | ✓ |
| [Tempor *et al.*, 2009] | ✓ | ✗ | ✗ |
| [Amet *et al.*, 2017] | ✓ | ✓ | ✗ |
| **Ours** | ✓ | ✓ | ✓ |

Table 1 A comparison of placeholder approaches. Checkmarks are placeholders too.

# Chapter III

# A Placeholder Contribution

This chapter presents the contribution of this thesis, and is the one that demonstrates most of the environments provided by the template.

It answers Problem 1, stated in Chapter I.

III.1 Method ..... 10
III.2 Results ..... 10
III.3 Discussion ..... 11

## III.1 Method

We look for a placeholder model $f_\theta$ minimizing the objective of Equation 1, where $\lambda > 0$ controls the strength of the placeholder regularizer.

$$\mathcal{L}(\theta) = \underbrace{\frac{1}{N}\sum_{i=1}^{N} \|f_\theta(\boldsymbol{x}_i) - \boldsymbol{y}_i\|_2^2}_{\text{placeholder fidelity}} + \lambda \underbrace{\|\theta\|_2^2}_{\text{placeholder regularizer}} \tag{1}$$

**Proposition 1** Existence of a placeholder minimizer

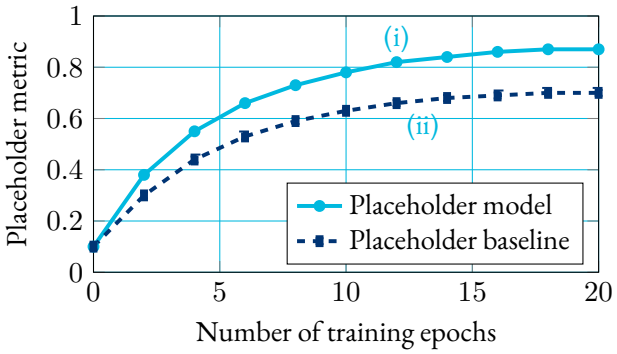
For every $\lambda > 0$, the objective of Equation 1 admits at least one minimizer $\theta^\star$.

The proof of Proposition 1 is deferred to Appendix A, so as not to interrupt the exposition. An unnumbered equation is obtained with the starred variant, which is convenient for intermediate steps:

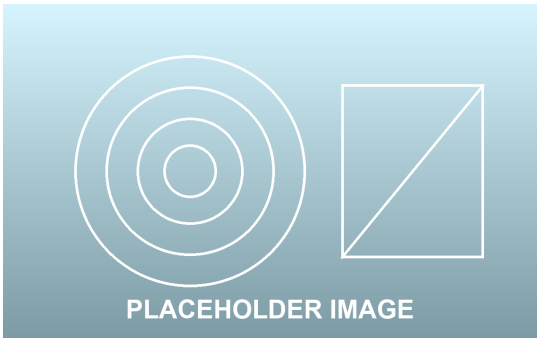
$$\nabla_\theta \mathcal{L}(\theta) = \frac{2}{N}\sum_{i=1}^{N} \left(f_\theta(\boldsymbol{x}_i) - \boldsymbol{y}_i\right) \nabla_\theta f_\theta(\boldsymbol{x}_i) + 2\lambda\theta$$

## III.2 Results

We evaluate the method of Section III.1 on placeholder datasets. Figure 2 reports the learning curves, and Table 2 the final scores.




(a) Placeholder metric along training, for our model and the baseline of [Amet *et al.*, 2017].




(b) A qualitative placeholder example.

Figure 2 Placeholder results. (a) reports the metric along training, while (b) shows a qualitative example.

Excepteur sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt mollit anim id est laborum. These results were partly published in [Lastname *et al.*, 2021] and [Coauthor *et al.*, 2024].

| Dataset | [Amet *et al.*, 2017] | Ours |
|---|---|---|
| Placeholder-A | $0.70 \pm 0.02$ | $\mathbf{0.87 \pm 0.01}$ |
| Placeholder-B | $0.61 \pm 0.03$ | $\mathbf{0.79 \pm 0.02}$ |
| Placeholder-C | $0.55 \pm 0.04$ | $\mathbf{0.68 \pm 0.03}$ |

Table 2 Final placeholder scores, averaged over five placeholder runs. Best results in bold.

## III.3 Discussion

**Limitations**

An unnumbered box, obtained with `boxenv*`, is handy for remarks that do not need to be cross-referenced. The placeholder method has placeholder limitations, which are discussed in Chapter IV.

# Chapter IV

# Conclusion & Perspectives

A short closing chapter, summarizing the contribution of Chapter III and listing the directions it opens.

IV.1 Summary ..... 14
IV.2 Perspectives ..... 14

## IV.1 Summary

This thesis addressed Problem 1, introduced in Chapter I. After the review of Chapter II, Chapter III proposed a placeholder method, whose objective is given in Equation 1 and whose results are reported in Table 2.

## IV.2 Perspectives

- Extending the placeholder method to placeholder data of higher dimension.
- Reducing the GPU cost of placeholder training, so that the approach scales beyond placeholder datasets.
- Studying the interpretability of the placeholder representations of Definition 1.

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua.

# Appendix A

# Proofs

Deferring proofs to an appendix keeps the main text readable.

The `proof` environment takes the label of the statement it proves, and displays a clickable reference to it in its title.

A.1 Existence of a Placeholder Minimizer . . . . . 16
A.2 A Proof Without a Statement . . . . . 16

## A.1 Existence of a Placeholder Minimizer

**Proof of Proposition 1**

The objective of Equation 1 is continuous in $\theta$, and coercive as soon as $\lambda > 0$, since $\|\theta\|_2^2 \to \infty$ when $\|\theta\|_2 \to \infty$. A continuous coercive function on $\mathbb{R}^d$ attains its infimum, hence the existence of a minimizer $\theta^\star$. □

## A.2 A Proof Without a Statement

The mandatory argument of `proof` can also be left empty, in which case the box is simply titled *"Proof"*.

**Proof**

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. □

# Bibliography

## References I Contributed To


[ Lastname *et al.*, 2021 ]
10

Firstname Lastname, Second Coauthor, Third Contributor (2021). **"Sed Do Eiusmod Tempor: Learning Placeholder Representations with CNNs"**. *Journal of Placeholder Research (vol. 12, nb. 3, pp. 42–58).*

[ Coauthor *et al.*, 2024 ]
10

Second Coauthor, Firstname Lastname (2024). **"Ut Enim Ad Minim Veniam: Scaling Placeholder Models on a Single GPU"**. *Transactions on Placeholder Systems (vol. 7, pp. 1–21).*


## Other References


[ Amet *et al.*, 2017 ]
— 6, 8, 10, 11

Consectetur Amet, Adipiscing Elit (2017). **"Seminal Placeholder Architectures for Structured Data"**. *31st Conference on Placeholder Systems (pp. 5998–6008).*

[ Ipsum *et al.*, 1998 ]
— 6, 8

Lorem Ipsum, Sit Dolor (1998). **"A Foundational Placeholder Study of Nothing in Particular"**. *Proceedings of the Placeholder Society (vol. 3, nb. 2, pp. 17–29).*

[ Tempor *et al.*, 2009 ]
— 6, 8

Eiusmod Tempor, Ut Labore (2009). **"The Elements of Placeholder Learning"**. *Placeholder Academic Publishing.*


## Online Resources


[ GitHub ]
August 19th, 2026

Bastien Pasdeloup. **"A LaTeX Template for HDR and PhD Manuscripts"**. *GitHub.*

[ Placeholder ]
August 19th, 2026

**"A Placeholder Online Resource"**. *Placeholder Organization.*

# Acronyms

| | |
|---|---|
| CNN | **C**onvolutional **N**eural **N**etwork |
| GPU | **G**raphics **P**rocessing **U**nit |
| HDR | **H**abilitation à **D**iriger des **R**echerches |
| ML | **M**achine **L**earning |
| PhD | **Ph**ilosophiæ **D**octor |
| SOTA | **S**tate **O**f **T**he **A**rt |

This text appears on the back cover, spread over two columns. Replace it with the abstract of your thesis. Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute irure dolor in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt mollit anim id est laborum. Sed ut perspiciatis unde omnis iste natus error sit voluptatem accusantium doloremque laudantium, totam rem aperiam, eaque ipsa quae ab illo inventore veritatis et quasi architecto beatae vitae dicta sunt explicabo.